

### 1.1.1

'the': 34392,  
'and': 24581,  
'to': 15159,  
'a': 10956,  
'he': 9618,  
'of': 9241,  
'was': 7858,  
'in': 6552,  
'his': 5785,  
'that': 5705

### 1.1.2

1. Punctuation remains attached to words, so the same word can be treated as multiple distinct tokens. For example, "said" and "said." are two different tokens.

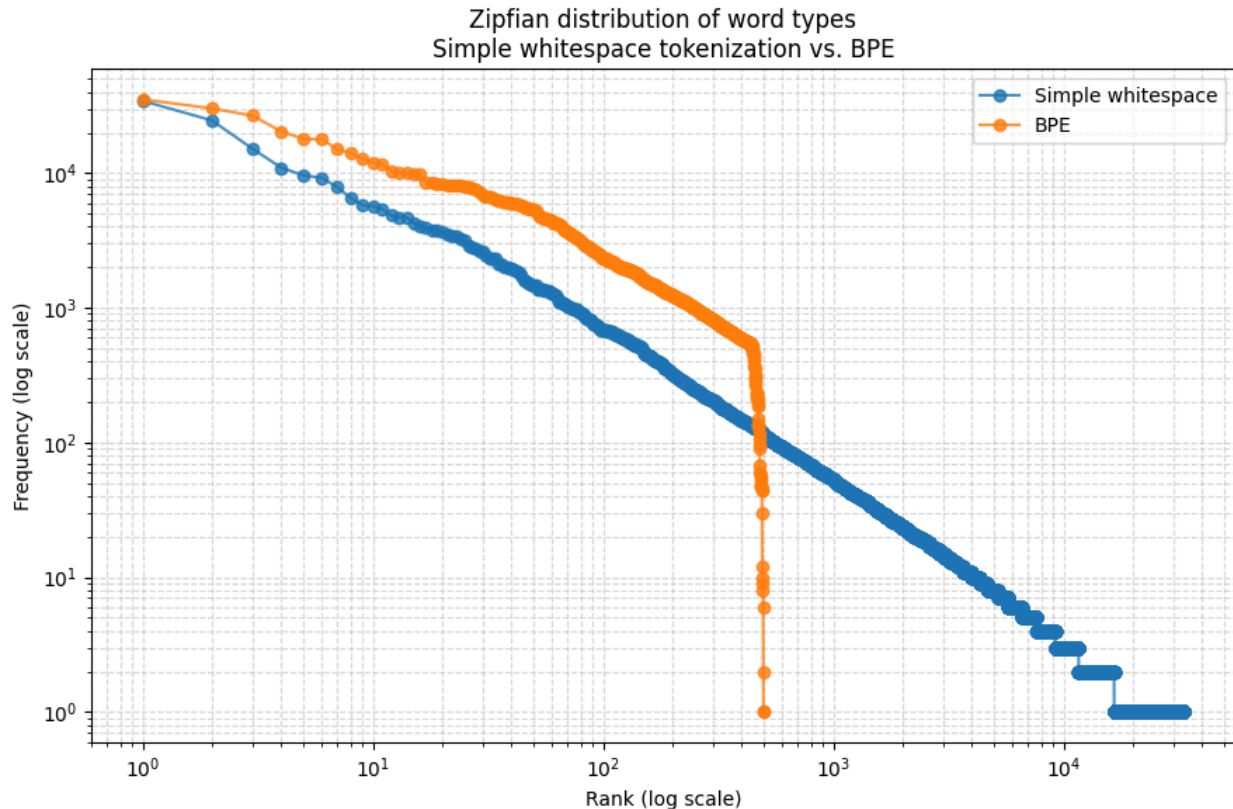
2. It produces a large vocabulary. For this 10,000-line corpus, it produced 33,132 tokens as it treated different forms of the same word as separate tokens. For example, "run", "runs", and "running".

### 1.2.1

Exclude the literal ``"`` boundary token

'the': 35285,  
',:': 30195,  
'and': 26701,  
's': 20350,  
'a': 17928,  
'to': 17864,  
'!': 15017,  
't': 14150,  
'he': 12738,  
'in': 11956,

### 1.2.2



Both curves are roughly straight (log-log) then curve down. BPE (orange) cuts off around rank 500. Simple whitespace (blue) extends to tens of thousands of ranks with a long tail down to frequency. BPE has a higher frequency than simple whitespace for most ranks.

### 1.2.3

BPE (orange) cuts off around rank 500 because the vocab\_size is 500. It can't have more than ~500 token types, so the curve must end there. The vertical-looking cluster near rank 500 is because tokens near the end of the ranking have very low frequencies('ö': 8,

'ä': 6,

'X': 2,

'/': 1,

'&': 1).

Simple whitespace tokenization has an unbounded vocabulary, which is why it stretches the tail way out.

BPE has a higher frequency because BPE splits words into smaller subword tokens, which can be shared across many different words.

### 2.2.1

Bag of words: highest-probability negative sample = ' the' (id=259),  $P=0.0657$

Neighboring tokens, N=1: highest-probability negative sample = ' the' (id=259),  $P=0.0654$

Neighboring tokens, N=2: highest-probability negative sample = ' the' (id=259),  $P=0.0655$

### 2.3.1

three examples of pairs of similar wordtypes that make sense

1. She - he 0.54
2. Very - well 0.49
3. Time - early 0.48

three that don't

1. Ġwant - 'll 0.47
2. If - oh 0.44
3. Ġcried - Ġreplied 0.48

A high similarity score between two seemingly unrelated words might be due to Skip-gram capturing distributional (syntactic) similarity, not semantic similarity. Function words and interjections often share very generic, high-frequency neighboring contexts even though they serve totally different meanings. “Ġcried” - “Ġreplied” fit this pattern well.

### 3.1.1

Solve correctly:

good:better::bad:? -> worse (expected: worse)  
big:bigger::small:? -> smaller (expected: smaller)  
dog:puppy::cat:? -> kitty (expected: kitty)  
paris:france::berlin:? -> Germany (expected: germany)

Solve wrong

king:queen::man:? -> lady (expected: woman)  
car:driver::train:? -> trains (expected: conductor)  
(if case sensitive: paris:france::berlin:? -> Germany (expected: germany))  
mouse:mice::goose:? -> rats (expected: geese)

### 3.2.1

|         |                 |                    |
|---------|-----------------|--------------------|
| penguin | -> fr: éléphant | -> back: elephant  |
| penguin | -> ja: 海        | -> back: ocean     |
| cheese  | -> fr: fromage  | -> back: sausage   |
| cheese  | -> ja: 美味しい     | -> back: delicious |
| sofa    | -> fr: canapé   | -> back: sofa      |
| sofa    | -> ja: ソファ      | -> back: bed       |
| jacket  | -> fr: pantalon | -> back: sleeves   |
| jacket  | -> ja: 上着       | -> back: clothing  |
| website | -> fr: internet | -> back: internet  |
| website | -> ja: インターネット  | -> back: internet  |

I found that only one got back to the original English term. For example, sofa -> fr: canapé -> back: sofa.

Most of them are converting into related meaning. For example, jacket -> ja: 上着 -> back: clothing, and penguin -> ja: 海 -> back: ocean.

Some of them are converting into similar meaning. For example, website -> fr: internet -> back: internet

#### 4.1.1

say: 204

go: 96

have: 94

cry: 71

come: 68

see: 54

look: 52

give: 43

think: 42

know: 42

#### 4.1.2

say (total occurrences: 204):

he: 23

Miller: 17

Prince: 15

Rocket: 15

Hans: 13

I: 11

Swallow: 9

Nightingale: 5

Giant: 5

rat: 5

Linnet: 5

Duck: 4

boy: 3

they: 3

Student: 3

Candle: 3

Professor: 2

.....(detail can see in the output in hw2.ipynb)

Stick: 1

it: 1

go (total occurrences: 96):

I: 22

he: 20  
you: 7  
they: 6  
everybody: 3  
she: 3  
it: 2

.....

we: 1  
Bang: 1  
!: 1  
minister: 1

have (total occurrences: 94):

I: 25  
he: 20  
she: 9  
you: 9  
they: 3  
we: 3  
who: 2  
mother: 1  
man: 1  
it: 1  
nephew: 1  
child: 1  
Hans: 1  
friend: 1  
story: 1  
anybody: 1  
none: 1  
everybody: 1

cry (total occurrences: 71):

he: 14  
Hans: 8  
they: 5  
she: 4  
Nightingale: 3  
tree: 3  
Rocket: 3  
who: 2  
Tree: 2  
press: 2  
Miller: 2

.....

Squib: 1  
order: 1  
Candle: 1  
Light: 1  
balloon: 1  
Frog: 1  
creature: 1  
Hallo: 1  
heaven: 1

come (total occurrences: 68):

you: 5  
Spring: 5  
they: 4  
he: 4  
Miller: 3  
I: 2  
snow: 2  
Day: 2

.....

Hail: 1  
perfume: 1  
bird: 1  
child: 1  
winter: 1  
your: 1  
sound: 1  
one: 1  
workman: 1

#### 4.1.3

There is a clear relationship between verbs and subject types. “Say,” “cry,” and “have” mostly take animate subjects, such as “he” and “I,” while “come” frequently takes inanimate subjects like “Spring,” “snow,” and “winter.” Interestingly, nearly all 204 occurrences of “say” have animate or personified subjects. Even objects like “Rocket” and “Tree” are treated as speaking characters in fairy tales. This reinforces the idea that communication verbs require sentient subjects, whereas “come” can describe the arrival of people, seasons, weather, or sounds without requiring agency.